Graph Theory

What is a graph?

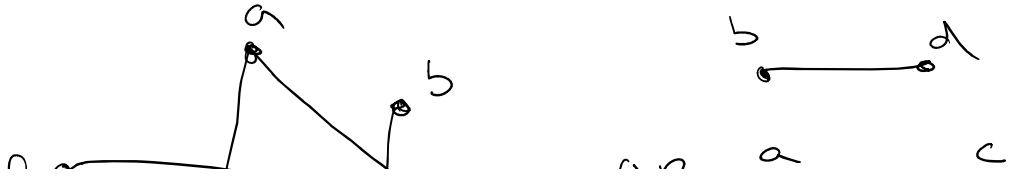
A graph G is a tuple, i.e., $G := (V, E)$, where V (or $V(G)$) is a set of elements called the vertex set (or node set) (each element in this set is called a vertex (or node)) and E (or $E(G)$) is a collection of 2-sized subsets of V and is called the edge set (each element in E is called an edge).

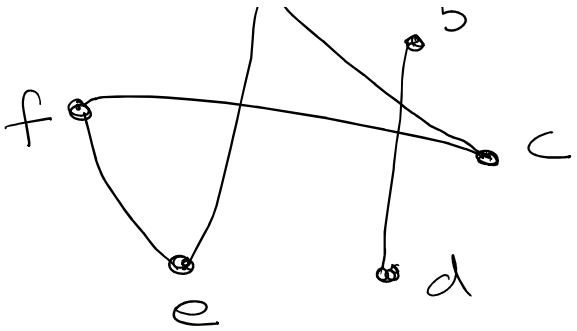
Example

$G = (V, E)$, where

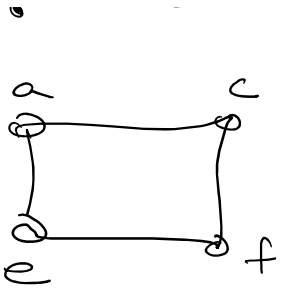
$$V = \{a, b, c, d, e, f\}$$

$$E = \{\{a, c\}, \{a, e\}, \{b, d\}, \{e, f\}, \{c, f\}\}$$





OR



A more general notion

Hypergraph → is a tuple

$H = (V, E)$

V → vertex set (a collection of elements)

E → a collection of ~~2-sized~~ subsets of V.

($\hat{E}$ a subset in E is called a hyperedge)

Example $H = (V, E)$

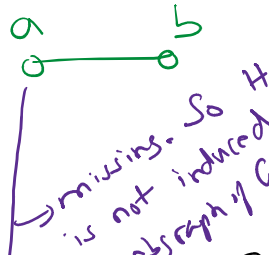
$V = \{a, b, c, d, e, f\}$

$E = \{\{a, b, c, d\}, \{c, e, f\}, \{a, e, f\}, \{b, d\}\}$

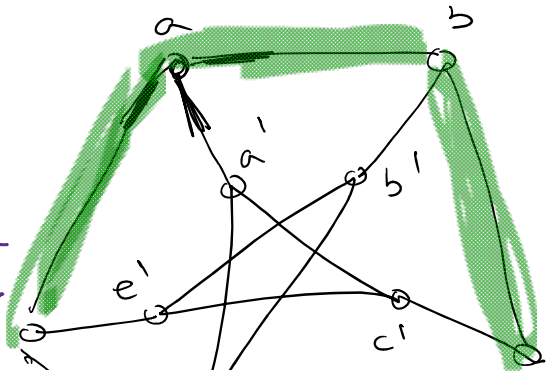
Some definitions on graphs

$G = (V, E)$

$H = (V', E')$

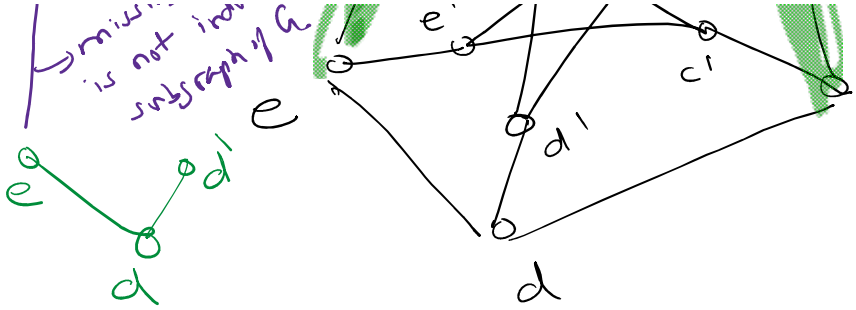


↳ misses. So H is not induced subgraph of G



$V = \{a, b, c, d, e, a', b', c', d', e'\}$

$E = \{\{a, b\}, \{b, c\}, \{c, d\}$



$E = \{ \{a,b\}, \{b,c\}, \{c,d\}, \{d,e\}, \{e,a\}, \{a,a'\}, \{b,b'\}, \{e,e'\}, \{a',c'\}, \{c',e'\}, \{e',b'\}, \{b',d'\}, \{d',a'\} \}$

OR (a,b)

OR

OR

$|V| = 10$

$|E| = 15$

a is a neighbour of b

$N_G(a) = \{b, a', e\}$

Notions

(i) degree of a vertex $\longrightarrow$ no. of edges it is present in.

$$\deg_G(a) = 3 \quad , \quad \deg_G(b) = 3$$

(ii) maximum degree of G, denoted by $\Delta(G)$, is

$$\Delta(G) := \max\{ \deg_G(v) : v \in V(G) \}$$

above example, $\Delta(G) = 3$.

(iii) minimum degree of G, denoted by

$$\delta(G) := \min\{ \deg_G(v) : v \in V(G) \}$$

above example, $\delta(G) = 3$

(iv) given two vertices u and v, we say u is adjacent to v if $\{u,v\} \in E(G)$.

u is a neighbor of v in G if $\{u,v\} \in E(G)$.

(v) Neighborhood of a vertex v in G is denoted $N_G(v)$ and is defined as $N_G(v) = \{u : \{u,v\} \in E(G)\}$

(vi) $H = (V', E')$ is a subgraph of $G = (V, E)$ if $V' \subseteq V$ and $E' \subseteq E$.

(vii) a vertex v is incident with an edge e if $v \in e$.
↳ 2-sized subset of V.

(ix) $H = (V', E')$ is an induced subgraph of $G = (V, E)$ if $V' \subseteq V$ and

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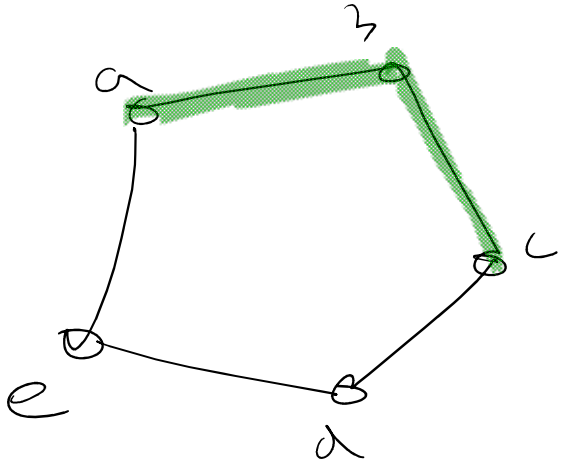
$E' = \{ (u,v) \in E(G) : u, v \in V' \}$

(x) $H = (V', E')$ is a spanning subgraph of $G = (V, E)$ if H is a subgraph of G and $V' = V$.

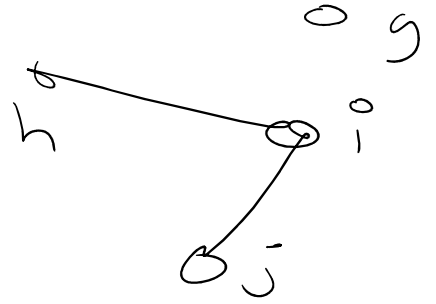
↳ spanning

(xi) Average degree of a graph G, denoted $d_{avg}(G)$ is

$$d_{avg}(G) := \frac{\sum_{v \in V(G)} deg_G(v)}{|V|}$$



for $H = (V, E)$





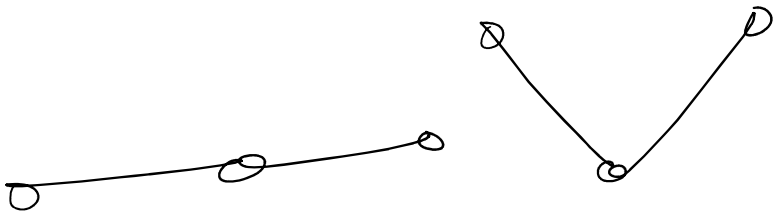


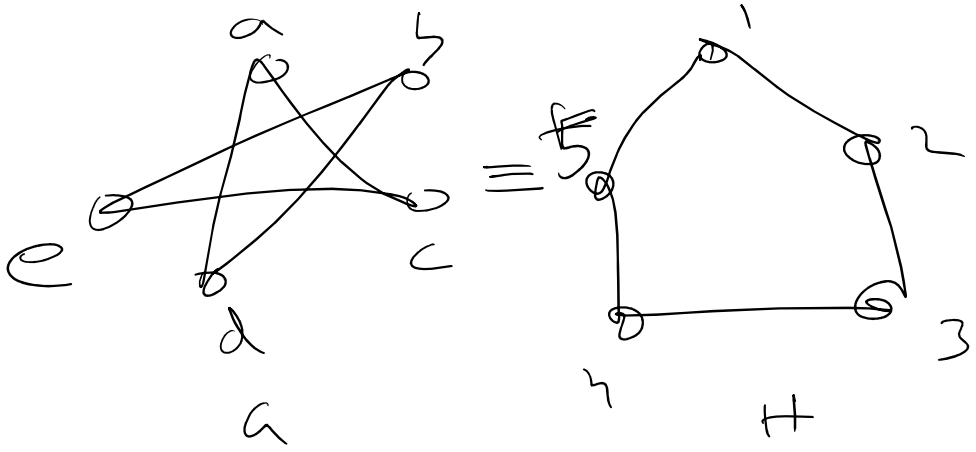
$V(H) = \{a, b, \ldots, l\}$

$E(H) = \{\{a,e\}, \{a,b\} \ldots \{k,l\}\}$

* sum of degrees $= 2 \cdot |E(G)|$.

Therefore no. of odd degree vertices is even.





G is isomorphic to H.

$f : V(G) \to V(G)$ as

$f(a) = 1$, $f(c) = 2$, $f(e) = 3$, $f(b) = 4$,

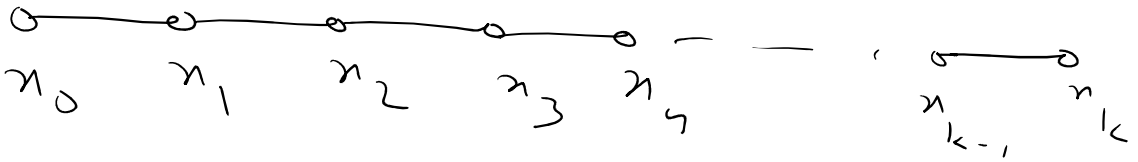
$f(d) = 5$

* A <u>path</u> is a non-empty graph

$P = (V, E)$, where

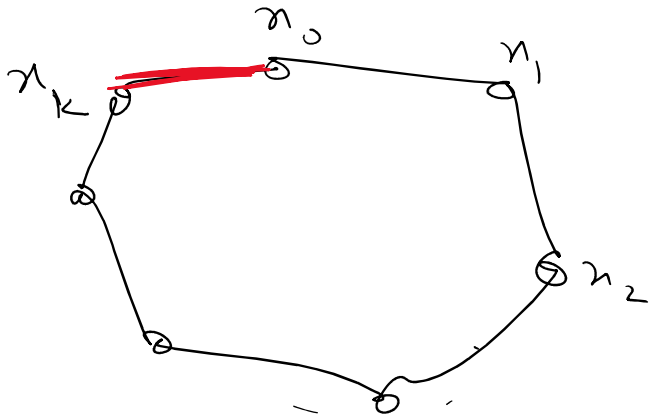
$$V = \{x_0, x_1, x_2, \ldots, x_k\}$$

$$E = \{\{x_0, x_1\}, \{x_1, x_2\}, \{x_2, x_3\}, \ldots \{x_{k-1}, x_k\}\}$$



<u>length</u> of a path is the no. of edges in it.

* Cycle



Let $P = (V, E)$ be a path, where $V = \{x_0, x_1, \ldots, x_k\}$

$$E = \{\{x_0, x_1\}, \{x_1, x_2\}, \ldots, \{x_{k-1}, x_k\}\}$$

Then, $P + \{x_k, x_0\}$ is a cycle

Lemma Every graph has (i) a path of length $\delta(G)$, and (ii) a cycle of length $\delta(G)+1$, $(\delta(G) \geq 2)$

(starting at a vertex with min degree)

Proof.

Proving (i) Consider a graph G with min degree $\delta(G)$. By mathematical induction on no. of vertices.

Base case: $|V(G)| = |V| = 1$.

$\delta(G) = 0$.

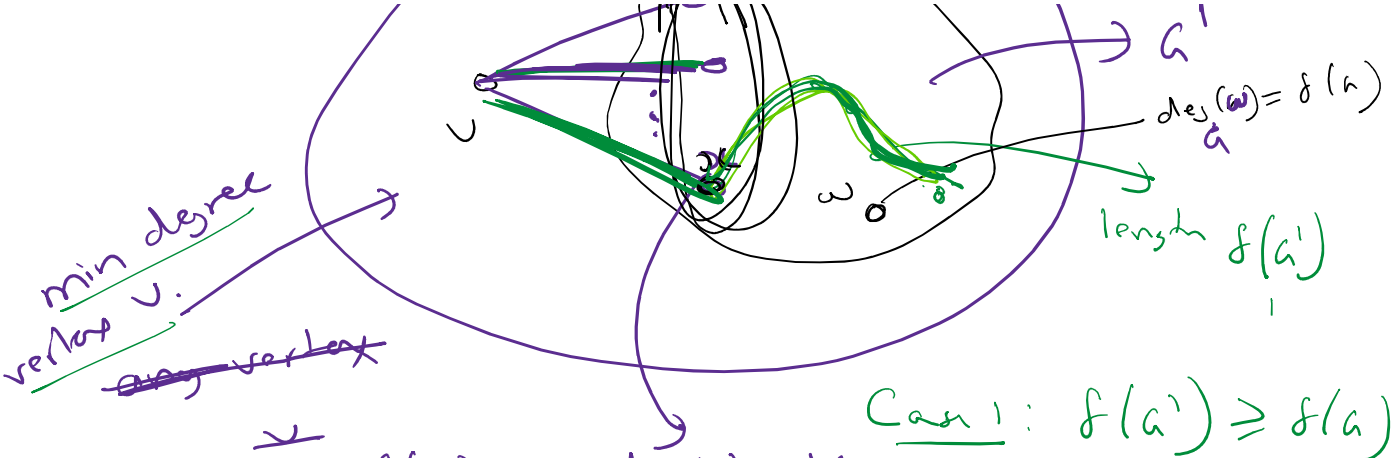
0-length path

Induction step. Assume the statement is TRUE for all graphs on n (at most) vertices.

Suppose G has $n+1$ vertices.



$\delta(G') = \delta(G)$



$\deg_G(v) = \delta(G)$.

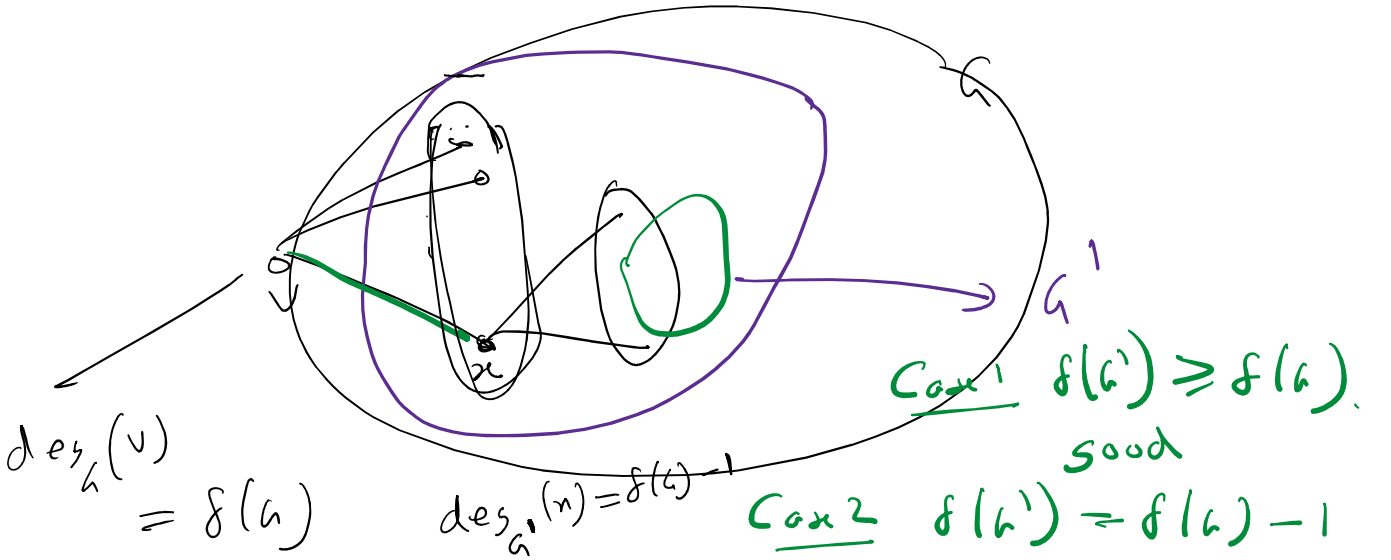
$\deg_G(x) = d(G)$

Case 1: $\delta(G') \geq \delta(G)$ good.

Case 2 $\delta(G') < \delta(G)$

$\deg_{G'}(x) = \delta(G) - 1$

Then, $\delta(G') = \delta(G) - 1$



$\deg_G(v) = \delta(G)$

Case 1 $\delta(G') \geq \delta(G)$ good

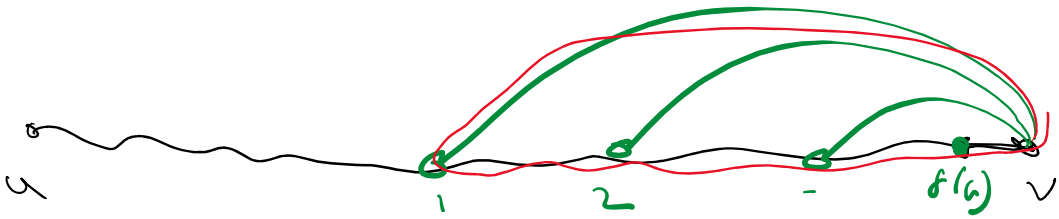
Case 2 $\delta(G') = \delta(G) - 1$

↳ How to go forward??

Alternate approach

Proving (i) and (ii)

Take the longest path in G.



$deg_G(v) \geq \delta(G)$

